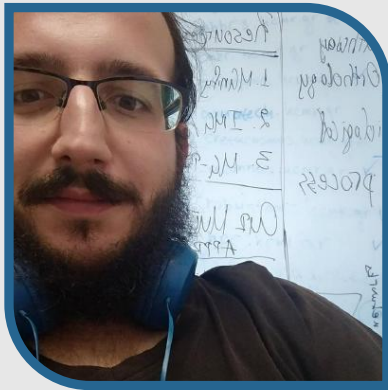




Zafeiropoulos Haris

Bioinformatics, Systems Biology, Microbial Ecology

-  April 24, 1991
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-  <https://hariszaf.github.io/>
-  haris.zafeiropoulos@kuleuven.be

About Me

I am developing bioinformatics approaches to enhance the study of microbial communities and their backbone: microbial interactions. As a bioinformatician, I have developed tools covering a range of fields: from metagenomics sequencing analysis to random flux sampling. My main scientific interests focus on microbial interactions and especially metabolic ones. I am keen on the ways microbial communities are instructed from their environment and the effect that this structure has back to the environment. In my endeavors, metabolic modeling has been a rather useful arsenal!

Social Network

-  Github Profile
-  [haris.zaf2](#)
-  [@haris_zaf](#)

Languages

-  English
-  French

Research projects - working experience

Formal Employment Projects

- | | | |
|------------------|---|--------------------|
| 2022 - currently | 3D'omics
Aim of the project is to come up with technology and protocols to generate 3D omics landscapes, achieving reconstructions of intestinal host microbiota ecosystems. My role on the project is the analysis of microbial networks integrating spatial and metabolic data to predict species interactions and/or influence of metabolites on species distributions using co-occurrence network analysis and Genome-Scale Metabolic Modeling. | Post-doc |
| 2022 - 2022 | A workflow for marine Genomic Observatories data analysis
An EOSC-Life-supported project aimed at making the large volumes of data produced by genomic observatories more easily interpretable by providing the taxonomic inventories of each sample in a timely and non-technical format. I was the lead developer of the delivered workflow, and the Principal Investigator in the second half of the project. | Research associate |
| 2018 - 2021 | PREGO: Process, environment, organism (PREGO)
PREGO is a systems-biology approach to elucidate ecosystem function at the microbial dimension. PREGO was the main project of my PhD and my role was to develop the data integration parts of the platform. | PhD |
| 2019 - present | ELIXIR-GR
ELIXIR-GR is the Greek National Node of the ESFRI European RI ELIXIR, a distributed e-Infrastructure aiming at the construction of a sustainable European infrastructure for biological information. My role on the project was to develop containerized-versions of various software tools. | technical support |
| 2018 - 2020 | RECONNECT
RECONNECT is an Interreg V-B "Balkan-Mediterranean 2014-2020" project. It aims to develop strategies for sustainable management of Marine Protected Areas (MPAs) and Natura 2000 sites. | PhD |

Unofficial Contributions

- | | | |
|----------------|---|-----|
| 2021 - 2023 | EMBRC
The European Marine Biological Resource Centre (EMBRC) was established in 2013 to advance fundamental and applied marine biology and ecology research, while promoting sustainable blue economy. Over the last years, I have contributed in tasks of the Greek node in the development of workflows for the analysis of the data produced by the network. | PhD |
| 2020 - ongoing | LifeWatch-ERIC
LifeWatch ERIC is a European Research Infrastructure Consortium providing e-Science research facilities to scientists investigating biodiversity and ecosystem functions and services in order to support society in addressing key planetary challenges. Since PEMA was selected to be part of an Internal Joint Initiative, I have been maintaining the workflow to address the needs of the infrastructure. | PhD |

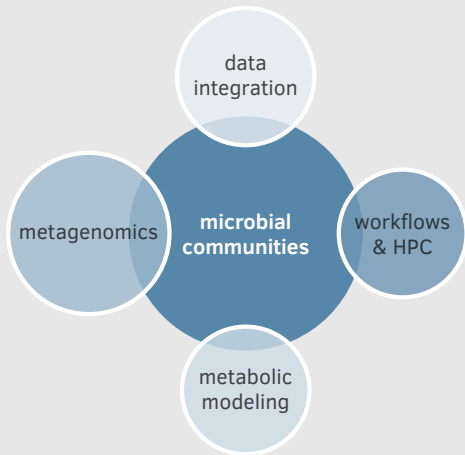
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Software

pema prego darn dingo
microbetag

Research fields



Memberships



Mikrobiokosmos



International Society of Microbial Ecology



Belgian Society for Microbiology

Education

Graduate studies

- 2018 – 2022 **PhD in Bioinformatics** University of Crete, Biology department
Dissertation IMBBC - HCMR
Microbial communities through the lens of high throughput sequencing, data integration and metabolic networks analysis
- 2016 – 2018 **M.Sc. in Bioinformatics** University of Crete, School of Medicine
grade: 9.1/10.0
Master Thesis IMBBC - HCMR
eDNA metabarcoding for biodiversity assessment: Algorithm design and bioinformatics analysis pipeline implementation

Undergraduate studies

- 2011 – 2016 **B.Sc. in Biology** National & Kapodistrian University of Athens
grade: 6.2 / 10.0
Bachelor Thesis School of Science, department of Biology
Morphology, morphometry and anatomy of species of the genus *Pseudamnicola* in Greece

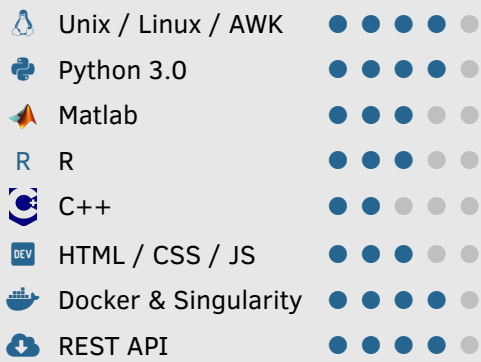
Publications

- 2025 **A Long Term Ecological Research Data Set From the Marine Genetic Monitoring Program ARMS MBON 2018 2020**
Daraghme N., Exter K., Pagnier J., ..., Zafeiropoulos, H., Zbawicka M., Pavloudi C., Obst M.
Molecular Ecology Resources, DOI: 10.1111/1755-0998.14073
- 2024 **Improving genome-scale metabolic models of incomplete genomes with deep learning**
Boer M.D., Melkonian C., Zafeiropoulos, H., Haas A.F., Garza D.R., Dutilh B.E.
iScience, DOI: 10.1016/j.isci.2024.111349
- 2024 **dingo: a Python package for metabolic flux sampling**
Chalkis, A., Fisikopoulos, V., Tsigaridas, E. and Zafeiropoulos, H.
Bioinformatics Advances, DOI: 10.1093/bioadv/vbae037
- 2024 **Predicting microbial interactions with approaches based on flux balance analysis: an evaluation.**
Joseph, C., Zafeiropoulos, H., Bernaerts, K. and Faust, K.
BMC Bioinformatics, DOI: 10.1186/s12859-024-05651-7
- 2024 **Establishing the ELIXIR Microbiome Community**
Finn, R.D., Balech, B., Burgin, J., Zafeiropoulos, H., Willassen N.P., Pelletier E., Batut B and 18 more
F1000Research, DOI: 10.12688/f1000research.144515.1
- 2023 **metaGOflow: a workflow for the analysis of marine Genomic Observatories shotgun metagenomics data**
Zafeiropoulos, H., Beracochea, M., Ninidakis, S., Exter, K., Potirakis, A., De Moro, G., Richardson, L., Corre, E., Machado, J., Pafilis, E. and Kotoulas, G.
GigaScience, DOI: 10.1093/gigascience/giad078
- 2023 **A pile of pipelines: An overview of the bioinformatics software for metabarcoding data analyses.**
Hakimzadeh, A., Abdala Asbun, A., ..., Zafeiropoulos, H., Anslan S.
Molecular Ecology Resources, DOI: 10.1111/1755-0998.13847

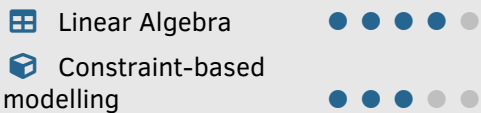
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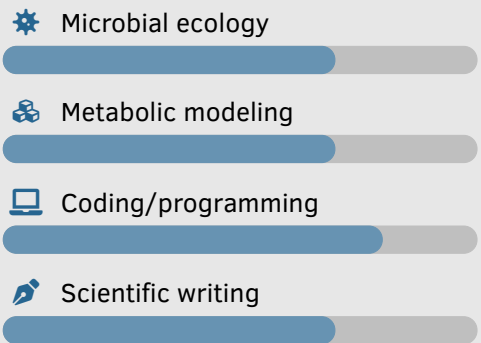
Coding Skills



Math and Stats



Barskills



Year	Publication Title	Authors	Journal	DOI
2023	Metabolic models of human gut microbiota: Advances and challenges.	Garza, D.R., Gonze, D., Zafeiropoulos, H., Liu, B. and Faust, K.	Cell Systems	DOI: 10.1016/j.cels.2022.11.002
2022	Deciphering the community structure and the functional potential of a hypersaline marsh microbial mat community.	Pavloudi, C. and Zafeiropoulos, H.	FEMS Microbiology Ecology	DOI: 10.1093/femsec/fiac141
2022	Automating the Curation Process of Historical Literature on Marine Biodiversity Using Text Mining: The DECO Workflow.	Paragkamian, S., Sarafidou, G., Mavraki, D., Pavloudi, C., Beja, J., Eliezer, M., Lipizer, M., Boicenco, L., Vandepitte, L., Perez-Perez, R., Zafeiropoulos, H. et al.	Frontiers in Marine Science	DOI: 10.3389/fmars.2022.940844
2022	PREGO: A Literature and Data-Mining Resource to Associate Microorganisms, Biological Processes, and Environment Types	Zafeiropoulos, H., Paragkamian, S., Ninidakis, S., Pavlopoulos, G.A., Jensen, L.J., Pafilis, E.	Microorganisms	DOI: 10.3390/microorganisms10020293
2021	The Dark mATteR iNvestigator (DARN) tool: getting to know the known unknowns in COI amplicon data.	Zafeiropoulos, H., Gargan, L., Hintikka, S., Pavloudi, C. & Carlsson, J.	Metabarcoding and Metagenomics	DOI:10.3897/mbmg.5.69657
2021	0s & 1s in marine molecular research: a regional HPC perspective	Zafeiropoulos, H., Gioti A., Ninidakis S., Potirakis A., ..., & Pafilis E.	GigaScience	DOI: 10.1093/gigascience/giab053
2021	Geometric Algorithms for Sampling the Flux Space of Metabolic Networks	Chalkis, A., Fisikopoulos, V., Tsigaridas, E. & Zafeiropoulos, H.	37th International Symposium on Computational Geometry (SoCG 2021)	DOI: 10.4230/LIPIcs.SocG.2021.21
2021	The Santorini Volcanic Complex as a Valuable Source of Enzymes for Bioenergy	Polymenakou, P.N., Nomikou, P., Zafeiropoulos, H., Mandalakis, M., Anastasiou, T.I., Kiliass, S., Kyrpides, N.C., Kotoulas, G. & Magoulas, A.	Energies	DOI: 10.3390/en14051414
2020	PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S ribosomal RNA, ITS & COI marker genes	Zafeiropoulos, H., Viet, H.Q., Vasileiadou, K., Potirakis, A., Arvanitidis, C., Topalis, P., Pavloudi, C. & Pafilis, E	GigaScience	DOI:10.1093/gigascience/giaa022
2024	microbetag: simplifying microbial network interpretation through annotation, enrichment tests & metabolic complementarity analysis	Zafeiropoulos, H., Delopoulos, E.I.M., Erega, A., Geirnaert, A., Morris, J. and Faust, K.	Genome Biology	DOI: 10.1101/2024.10.01.616208

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Conferences

2024	19th International Symposium on Microbial Ecology @ Cape Town, South Africa <i>Poster presentation:</i> microbetag: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis	ISME19
2024	Belgian Society for Microbiology: Annual Symposium @ Brussels, Belgium <i>Poster presentation:</i> microbetag: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis	BSM2024
2022	1st Applied HoloGenomics conference @ Bilbao, Spain <i>Poster presentation:</i> microbetag: a microbial co-occurrence network annotator	AHC2022
2022	18th International Symposium on Microbial Ecology @ Laussane, Switzerland <i>Poster presentation:</i> Reverse ecology and systems biology approaches to identify key processes ensuring life in extreme environments	ISME18
2021	Bioinformatics Open Source Conference (BOSC) - online <i>Flash talk:</i> dingo: A python library for metabolic networks sampling & analysis.)	BOSC2021
2021	1st DNAQUA International Conference - online <i>Flash talk:</i> PEMA v2: addressing metabarcoding bioinformatics analysis challenges	DNAQUA
2020	Federation of European Microbiological Societies (FEMS) - online <i>Flash talk:</i> Mining literature and -omics (meta)data to associate microorganisms, biological processes and environment types	FEMS2020
2020	PyData Global - online <i>Oral presentation:</i> Geometric and statistical methods in systems biology: the case of metabolic networks	PyData2020
2019	network2019: 4th Symposium on Ecological Networks @ Paris, France Participation.	network2019
2019	8th International Barcode of Life Conference @ Trondheim, Norway <i>ePoster presentation:</i> P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis	iBOL
2018	European Conference on Computational Biology (ECCB) 2018 @ Athens, Greece <i>Poster presentation:</i> P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis	ECCB18

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Workshops & Summer schools

2025	Metabolic models applied to microbiomes @ Antony, France - <i>Organizing committee</i> - Metabolic model analysis methods (theory & hands-on)	MMAM
2023	Summer School on Microbial Community Design and Control @ Leuven, Belgium Assisting on "metabolic network reconstruction" topic	DCMC
2023	Economic Principles in Cell Physiology @ Paris, France Mathematical modeling of cellular systems and "resource allocation thinking". <i>Poster presentation</i> : "sampler all": exploring potential flux sampling applications	EPCP2023
2022	Microbial communities: current approaches and open challenges @ Cambridge, UK Interdisciplinary exchanges in microbial communities' research and the emerging challenge areas.	UMCW06
2022	ELIXIR Fluxomics Training School 2021 An introduction to the field of fluxomics and the experimental and computational methods used to estimate and predict metabolic fluxes	ELIXFLUX
2020	Metagenomics, Metatranscript- omics and multi 'omics for microbial community studies A Physalia Course from Prof. Dr. Curtis Huttenhower	material

Supervision

2025-ongoing	Sourav Pattanaik, online internship student Populating μ GgrowrhDB	
2024-ongoing	Sotiris Touloupoulos, ERASMUS student Pre- and post-sampling features to leverage flux sampling at both the strain and the community level	GitHub repo
2024-ongoing	Xi Peng, Visiting PhD student Enhancing microbetag's functional annotations with quorum sensing prediction capabilities	
2024-ongoing	Anna Voukouna, MSc student @ Bioinformatics DUTH, Greece Investigation of microbial metabolic interactions potential across the Genome Taxonomy Database representative genomes	
2024	Sotiris Touloupoulos, Google Summer of Code 2024 with GeomScale org Pre- and post-sampling features to leverage flux sampling at both the strain and the community level	blogpost
2024-2025	Andrey Radev, MSc student @ Bioinformatics KU Leuven Visualisation and analysis of microbial growth data to facilitate interactions investigation and interactions' strength assessment	
2023-2024	Sofia Monsalve Duarte, MSc student @ Bioinformatics KU Leuven "Creation and implementation of a database interface for the collection and analysis of microbial growth data"	
2023-2024	Ermis Ioannis Michail Delopoulos, Google Summer of Code 2023 with NRNB org. Development of a Cytoscape App for microbe-microbe association networks	post
2022-2024	Ermis Ioannis Michail Delopoulos, MSc student @ Bioinformatics KU Leuven "Development of a graphical user interface for microbetag"	

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2022-2023 **Julia Casado Gómez-Pallete, MSc student @ Bioinformatics KU Leuven** thesis
"Development of a database for human gut bacterial growth curves"

Teaching

2025	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2025	Introduction in metabolic modeling GitHub repo Theoretical and practical session on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective (part II) slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace, Greece
2022	Genome-scale model reconstruction GColab notebook Lectures on "Master en bioinformatique et modélisation" of the Université Libre de Bruxelles.
2021	Microbiome Data Analyses Workshop Online MDAWO Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.
2020	Environmental DNA & DNA metabarcoding slides Lecture on MSc "Environmental Biology" in University of Crete.
2019	8th Conference of Hellenic Society of Mikrobiokosmos MKB8 Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.

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Awards & Scholarships

2021	9th conference of the Hellenic Society of Mikrobiokosmos MBK9 Oral presentation award: Sampling the flux space of microbial metabolic networks for (un)biased assessment of the possible functional states: the example of Sars-Cov-2 on the human alveolar macrophage metabolic network.
2021	Google Summer of Code GSoC Project title: <i>From DNA sequences to metabolic interactions: building a pipeline to extract key metabolic processes</i>
2021	European Molecular Biology Organization Short-Term Fellowship EMBO Project title: <i>"Exploiting data integration, text-mining and computational geometry to enhance microbial interactions inference from co-occurrence networks"</i>
2020	Federation of European Microbiological Societies Meeting Attendance Grant FEMS for joining the <i>"Metagenomics, Metatranscript- omics and multi 'omics for microbial community studies"</i> Physalia course
2019	Short Term Scientific Mission (STSM) @ DNAqua-net COST action DNAqua-net Project title <i>A comparison of bioinformatic pipelines and sampling techniques to enable benchmarking of DNA metabarcoding</i> . Report.
2018	Best Poster Award @ Hellenic Bioinformatics conference HBIO for <i>PEMA: a Pipeline for Environmental DNA Metabarcoding Analysis</i>

References

Karoline Faust	Department of Microbiology, Immunology and Transplantation, Rega Institute for Medical Research, Laboratory of Molecular Bacteriology, KU Leuven Leuven, 3000, Leuven, Belgium ☎: +3216322698 @: karoline.faust@kuleuven.be
Pafilis Evangelos	Institute of Marine Biology, Biotechnology & Aquaculture, Hellenic Centre for Marine Research, (IMBBC - HCMR), Gournes, Pediados, P.O. Box 2214, Heraklion Crete 71003, Greece ☎: +30 2810 337740 @: pafilis@hcmr.gr
Elias Tsigaridas	Institut de Mathématiques de Jussieu - Paris Rive Gauche, Sorbonne Université - Campus Pierre et Marie Curie, Case courrier 247, 4, place Jussieu, 75252, Paris Cedex 05, France ☎: +33 014427 8539 @: elias.tsigaridas@inria.fr
Georgios Kotoulas	Institute of Marine Biology, Biotechnology & Aquaculture, Hellenic Centre for Marine Research, (IMBBC - HCMR) Gournes, Pediados, P.O. Box 2214, Heraklion Crete 71003, Greece ☎: +30 2810 337740 @: kotoulas@hcmr.gr